

Anisotropy of thermal conductivity — Solution

X23 (2023) — English translation

A1

0.60 pts

Using the least squares method, obtain formulas for finding the coefficients A and B for each temperature value T_0 and for each material. Express your answer in terms of sample average values of products of the form $\Delta T^{k_1} d^{k_2}$, where $k_{1,2}$ are some numbers.

According to the method described in the condition, we compose a system of equations:

$$\begin{cases} \frac{\partial S}{\partial A} = 0 \\ \frac{\partial S}{\partial B} = 0 \end{cases}, \quad \text{where} \quad S = \sum_i (\Delta T_{i \text{ exp}} - \Delta T_{i \text{ th}}(d_i, A, B))^2 = \sum_i (\Delta T_i - Ad_i + BT_0^{1/2} d_i^{3/2})^2$$

Thus, the system will take the form:

$$\begin{cases} \frac{\partial}{\partial A} \sum_i (\Delta T_i - Ad_i + BT_0^{1/2} d_i^{3/2})^2 = 0 \\ \frac{\partial}{\partial B} \sum_i (\Delta T_i - Ad_i + BT_0^{1/2} d_i^{3/2})^2 = 0 \end{cases} \implies \begin{cases} 0 = \sum_i (\Delta T_i d_i - Ad_i^2 + BT_0^{1/2} d_i^{5/2}) \\ 0 = \sum_i (\Delta T_i d_i^{3/2} - Ad_i^{5/2} + BT_0^{1/2} d_i^3) \end{cases}$$

This is a system of linear equations for the coefficients A and B . The coefficients in it are necessary products of the form $\Delta T^{k_1} d^{k_2}$. Using the average values of these products over the experimental sample, this system is written somewhat more elegantly in the form:

$$\begin{cases} 0 = \overline{\Delta T d} - A \overline{d^2} + B T_0^{1/2} \overline{d^{5/2}} \\ 0 = \overline{\Delta T d^{3/2}} - A \overline{d^{5/2}} + B T_0^{1/2} \overline{d^3} \end{cases}$$

Solving this system, we finally obtain:

$$A = \frac{\overline{\Delta T d^{3/2}} \cdot \overline{d^{5/2}} - \overline{\Delta T d} \cdot \overline{d^3}}{\overline{d^{5/2}}^2 - \overline{d^2} \cdot \overline{d^3}}$$

$$B = \frac{1}{\sqrt{T_0}} \frac{\overline{\Delta T d^{3/2}} \cdot \overline{d^2} - \overline{\Delta T d} \cdot \overline{d^{5/2}}}{\overline{d^{5/2}}^2 - \overline{d^2} \cdot \overline{d^3}}$$

Answer:

$$A = \frac{\overline{\Delta T d^{3/2}} \cdot \overline{d^{5/2}} - \overline{\Delta T d} \cdot \overline{d^3}}{\overline{d^{5/2}}^2 - \overline{d^2} \cdot \overline{d^3}}$$

$$B = \frac{1}{\sqrt{T_0}} \frac{\overline{\Delta T d^{3/2}} \cdot \overline{d^2} - \overline{\Delta T d} \cdot \overline{d^{5/2}}}{\overline{d^{5/2}}^2 - \overline{d^2} \cdot \overline{d^3}}$$

A2

1.20 pts

For molybdenum sulfide (MoS_2), find A and B for each of the temperatures T_0 .

$T_0 = 273 \text{ K}$, $\overline{d}$, μm 20 40 60 80 100 120 140 160 180 200 ΔT , K 0.138 0.266 0.394 0.532 0.662 0.777 0.831 0.954 1.137 1.189 0.688 $\Delta T d$, K $\cdot \mu\text{m}$ 2.76 10.64 23.64 42.56 66.20 93.24 116.34 152.64 204.66 237.80 95.05 $\Delta T d^{3/2}$, K $\cdot \mu\text{m}^{3/2}$ 12.34 6729 18311 380.67 662.00 1021.39 1376.55 1930.76 2745.80 3363.00 1174.29 d^2 , μm^2 400 1600 3600 6400 10000 14400 19600 25600 32400 40000 15400 $d^{5/2}$, $\mu\text{m}^{5/2}$ 1788.85 10119.29 27885.48 57243.34 100000.00 157744.10 231910.33 323817.23 434691.61 565685.42 191088.57 d^3 , μm^3 8000

64000 216000 512000 1000000 1728000 2744000 4096000 5832000 8000000 2420000 $T_0 = 323 \text{ K}\bar{d}$, μm 20 40 60 80 100 120 140 160 180 200 ΔT , K 0.137 0.271 0.390 0.496 0.577 0.719 0.788 0.877 1.046 1.098 0.640 ΔTd , K $\cdot \mu\text{m}$ 2.74 10.84 23.40 39.68 57.70 86.28 110.32 140.32 188.28 219.60 87.92 $\Delta Td^{3/2}$, K $\cdot \mu\text{m}^{3/2}$ 12.25 68.56 181.26 354.91 577.00 945.15 1305.32 1774.92 2526.04 3105.61 1085.10 d^2 , μm^2 400 1600 3600 6400 10000 14400 19600 25600 32400 40000 15400 $d^{5/2}$, $\mu\text{m}^{5/2}$ 1788.85 10119.29 27885.48 57243.34 100000.00 157744.10 231910.33 323817.23 434691.61 565685.42 191088.57 d^3 , μm^3 8000 64000 216000 512000 1000000 1728000 2744000 4096000 5832000 8000000 2420000 $T_0 = 373 \text{ K}\bar{d}$, μm 20 40 60 80 100 120 140 160 180 200 ΔT , K 0.139 0.267 0.373 0.483 0.563 0.676 0.760 0.856 0.901 1.023 0.604 ΔTd , K $\cdot \mu\text{m}$ 2.78 10.68 22.38 38.64 56.30 81.12 106.40 136.96 162.18 204.60 82.20 $\Delta Td^{3/2}$, K $\cdot \mu\text{m}^{3/2}$ 12.43 67.55 173.35 345.61 563.00 888.63 1258.94 1732.42 2175.87 2893.48 1011.13 d^2 , μm^2 400 1600 3600 6400 10000 14400 19600 25600 32400 40000 15400 $d^{5/2}$, $\mu\text{m}^{5/2}$ 1788.85 10119.29 27885.48 57243.34 100000.00 157744.10 231910.33 323817.23 434691.61 565685.42 191088.57 d^3 , μm^3 8000 64000 216000 512000 1000000 1728000 2744000 4096000 5832000 8000000 2420000

Answer: T_0 , K 273 323 373 A, $\frac{\text{K}}{\text{mm}}$ 7.46 7.18 7.59 B, $\frac{\text{K}^{1/2}}{\text{mm}^{3/2}}$ 0.199 0.208 0.298

A3

0.50 pts

Find out whether it is possible to conclude from the results of the experiment that thermal conductivity $\kappa_{\perp}$ depends on temperature, underline the correct option on the answer sheet (both sides of the inequality must be explicitly calculated in the solution). Find the arithmetic mean $\bar{A}$ and calculate the thermal conductivity $\kappa_{\perp}$ from it.

We substitute the numbers (taking into account that $T_{0\text{max}} = 373 \text{ K}$, $d_{\text{max}} = 0.2 \text{ mm}$):

$$\Delta A < \bar{B}T_{0\text{max}}^{1/2}d_{\text{max}}^{1/2} \implies 0.41 \frac{\text{K}}{\text{mm}} < 2.03 \frac{\text{K}}{\text{mm}} - \text{depends} / \underline{\text{no depends}}.$$

Answer:

$$\bar{A} = 7.41 \frac{\text{K}}{\text{mm}} \kappa_{\perp} = 53.3 \frac{\text{mW}}{\text{m} \cdot \text{K}} \neq \kappa_{\perp}(T_0)$$

A4

1.70 pts

Follow steps **A2** and **A3** for tungsten sulfide (WS_2).

$T_0 = 273 \text{ K}$. $\bar{d}$, μm 20 40 60 80 100 120 140 160 180 200 ΔT , K 0.191 0.381 0.571 0.764 0.946 1.100 1.304 1.443 1.577 1.809 1.009 ΔTd , K $\cdot \mu\text{m}$ 3.82 15.24 34.26 61.12 94.60 132.00 182.56 230.88 283.86 361.80 140.01 $\Delta Td^{3/2}$, K $\cdot \mu\text{m}^{3/2}$ 17.08 96.39 265.38 546.67 946.00 1445.99 2160.08 2920.43 3808.38 5116.62 1732.30 d^2 , μm^2 400 1600 3600 6400 10000 14400 19600 25600 32400 40000 15400 $d^{5/2}$, $\mu\text{m}^{5/2}$ 1788.85 10119.29 27885.48 57243.34 100000.00 157744.10 231910.33 323817.23 434691.61 565685.42 191088.57 d^3 , μm^3 8000 64000 216000 512000 1000000 1728000 2744000 4096000 5832000 8000000 2420000 $T_0 = 323 \text{ K}\bar{d}$, μm 20 40 60 80 100 120 140 160 180 200 ΔT , K 0.195 0.384 0.566 0.727 0.907 1.083 1.202 1.354 1.518 1.701 0.964 ΔTd , K $\cdot \mu\text{m}$ 3.9 15.36 33.96 58.16 90.7 129.96 168.28 216.64 273.24 340.2 133.04 $\Delta Td^{3/2}$, K $\cdot \mu\text{m}^{3/2}$ 17.44 97.15 263.05 520.20 907.00 1423.64 1991.12 2740.30 3665.90 4811.15 1643.70 d^2 , μm^2 400 1600 3600 6400 10000 14400 19600 25600 32400 40000 15400 $d^{5/2}$, $\mu\text{m}^{5/2}$ 1788.85 10119.29 27885.48 57243.34 100000.00 157744.10 231910.33 323817.23 434691.61 565685.42 191088.57 d^3 , μm^3 8000 64000 216000 512000 1000000 1728000 2744000 4096000 5832000 8000000 2420000 $T_0 = 373 \text{ K}\bar{d}$, μm 20 40 60 80 100 120 140 160 180 200 ΔT , K 0.201 0.367 0.549 0.733 0.877 1.037 1.172 1.330 1.439 1.629 0.933 ΔTd , K $\cdot \mu\text{m}$ 4.02 14.68 32.94 58.64 87.70 124.44 164.08 212.80 259.02 325.80 128.41 $\Delta Td^{3/2}$, K $\cdot \mu\text{m}^{3/2}$ 17.98 92.84 255.15 524.49 877.00 1363.17 1941.42 2691.73 3475.12 4607.51 1584.64 d^2 , μm^2 400 1600 3600 6400 10000 14400 19600 25600 32400 40000 15400 $d^{5/2}$, $\mu\text{m}^{5/2}$ 1788.85 10119.29 27885.48 57243.34 100000.00 157744.10 231910.33 323817.23 434691.61 565685.42 191088.57 d^3 , μm^3 8000 64000 216000 512000 1000000 1728000 2744000 4096000 5832000 8000000 2420000 Substitute the numbers (taking into account that $T_{0\text{max}} = 373 \text{ K}$, $d_{\text{max}} = 0.2 \text{ mm}$):

$$\Delta A < \bar{B}T_{0\text{max}}^{1/2}d_{\text{max}}^{1/2} \implies 0.23 \frac{\text{K}}{\text{mm}} < 2.17 \frac{\text{K}}{\text{mm}} - \text{depends} / \underline{\text{no depends}}$$

We substitute the numbers (taking into account that $T_{0\max} = 373 \text{ K}$, $d_{\max} = 0.2 \text{ mm}$):

$$\Delta A < \overline{B} T_{0\max}^{1/2} d_{\max}^{1/2} \implies 0.23 \frac{\text{K}}{\text{mm}} < 2.17 \frac{\text{K}}{\text{mm}} - \text{depends} / \underline{\text{no depends}}$$

$$\overline{A} = 10.47 \frac{\text{K}}{\text{mm}} \kappa_{\perp} = 37.7 \frac{\text{mW}}{\text{m} \cdot \text{K}} \neq \kappa_{\perp}(T_0)$$

Answer: T_0 , K 273 323 373 A , $\frac{\text{K}}{\text{mm}}$ 10.37 10.44 10.60 B , $\frac{\text{K}^{1/2}}{\text{mm}^{3/2}}$ 0.197 0.256 0.300

$$\overline{A} = 10.47 \frac{\text{K}}{\text{mm}} \kappa_{\perp} = 37.7 \frac{\text{mW}}{\text{m} \cdot \text{K}} \neq \kappa_{\perp}(T_0)$$

B1

0.60 pts

Using the least squares method, obtain a formula for finding the coefficient A for each temperature value T_0 and for each material. Express your answer in terms of sample average values of products of the form $\Delta T^{k_1} d^{k_2}$, where $k_{1,2}$ are some numbers.

According to the method described in the condition, we compose a system of equations:

$$\begin{cases} \frac{\partial S}{\partial A} = 0 \\ \frac{\partial S}{\partial B} = 0 \end{cases}, \quad \text{where} \quad S = \sum_i (\Delta T_{i \text{ exp}} - \Delta T_{i \text{ th}}(d_i, A, B))^2 = \sum_i \left(\Delta T_i - \frac{A}{d_i} + \frac{B}{d_i^2} \right)^2$$

Thus, the system will take the form:

$$\begin{cases} \frac{\partial}{\partial A} \sum_i \left(\Delta T_i - \frac{A}{d_i} + \frac{B}{d_i^2} \right)^2 = 0 \\ \frac{\partial}{\partial B} \sum_i \left(\Delta T_i - \frac{A}{d_i} + \frac{B}{d_i^2} \right)^2 = 0 \end{cases} \implies \begin{cases} 0 = \sum_i (\Delta T_i d_i^{-1} - A d_i^{-2} + B d_i^{-3}) \\ 0 = \sum_i (\Delta T_i d_i^{-2} - A d_i^{-3} + B d_i^{-4}) \end{cases}$$

This is a system of linear equations for the coefficients A and B . The coefficients in it are necessary products of the form $\Delta T^{k_1} d^{k_2}$. Using the average values of these products over the experimental sample, this system is written somewhat more elegantly in the form:

$$\begin{cases} 0 = \overline{\Delta T d^{-1}} - A \overline{d^{-2}} + B \overline{d^{-3}} \\ 0 = \overline{\Delta T d^{-2}} - A \overline{d^{-3}} + B \overline{d^{-4}} \end{cases}$$

Solving this system for A , we finally obtain:

$$A = \frac{\overline{\Delta T d^{-2}} \cdot \overline{d^{-3}} - \overline{\Delta T d^{-1}} \cdot \overline{d^{-4}}}{\overline{d^{-3^2}} - \overline{d^{-2}} \cdot \overline{d^{-4}}}.$$

Answer:

$$A = \frac{\overline{\Delta T d^{-2}} \cdot \overline{d^{-3}} - \overline{\Delta T d^{-1}} \cdot \overline{d^{-4}}}{\overline{d^{-3^2}} - \overline{d^{-2}} \cdot \overline{d^{-4}}}$$

B2

1.00 pts

For molybdenum sulfide (MoS_2), find A for each of the temperatures T_0 . From here, for each T_0 , find the values of $\kappa_{\parallel}$. Find the thermal conductivity anisotropy coefficient $\rho = \frac{\kappa_{\parallel}}{\kappa_{\perp}}$ at temperature $T_0 = 273 \text{ K}$.

$T_0 = 273 \text{ K}$. $\overline{d}$, μm 120 140 160 180 200 ΔT , K 1.372 1.157 1.055 0.948 0.846 $\Delta T d^{-1}$, 10^{-3} K .

μm^{-1} 11.43 8.26 6.59 5.27 4.23 7.16 $\Delta T d^{-2}$, 10^{-6} K $\cdot \mu\text{m}^{-2}$ 95.28 59.03 41.21 29.26 21.15 49.19
 d^{-2} , 10^{-6} μm^{-2} 69.44 51.02 39.06 30.86 25.00 43.08 d^{-3} , 10^{-9} μm^{-3} 578.70 364.43 244.14 171.47
125.00 296.75 d^{-4} , 10^{-9} μm^{-4} 4.823 2.603 1.526 0.953 0.625 2.106 $T_0 = 323$ K $\bar{d}$, μm 120 140 160
180 200 ΔT , K 1.407 1.247 1.144 1.019 0.948 $\Delta T d^{-1}$, 10^{-3} K $\cdot \mu\text{m}^{-1}$ 11.73 8.91 7.15 5.66 4.74 7.64
 $\Delta T d^{-2}$, 10^{-6} K $\cdot \mu\text{m}^{-2}$ 97.71 63.62 44.69 31.45 23.70 52.23 d^{-2} , 10^{-6} μm^{-2} 69.44 51.02 39.06 30.86
25.00 43.08 d^{-3} , 10^{-9} μm^{-3} 578.70 364.43 244.14 171.47 125.00 296.75 d^{-4} , 10^{-9} μm^{-4} 4.823 2.603
1.526 0.953 0.625 2.106 $T_0 = 373$ K $\bar{d}$, μm 120 140 160 180 200 ΔT , K 1.455 1.356 1.194 1.107 1.024
 $\Delta T d^{-1}$, 10^{-3} K $\cdot \mu\text{m}^{-1}$ 12.13 9.69 7.46 6.15 5.12 8.11 $\Delta T d^{-2}$, 10^{-6} K $\cdot \mu\text{m}^{-2}$ 101.04 69.18 46.64 34.17
25.60 55.33 d^{-2} , 10^{-6} μm^{-2} 69.44 51.02 39.06 30.86 25.00 43.08 d^{-3} , 10^{-9} μm^{-3} 578.70 364.43 244.14
171.47 125.00 296.75 d^{-4} , 10^{-9} μm^{-4} 4.823 2.603 1.526 0.953 0.625 2.106

$$\rho = 9.85 \cdot 10^2$$

Answer: T_0 , K 273 323 373 A , $\mu\text{m} \cdot \text{K}$ 179.6 218.8 247.5 $\varkappa_{||}$, $\frac{\text{W}}{\text{m} \cdot \text{K}}$ 52.5 43.1 38.1

$$\rho = 9.85 \cdot 10^2$$

B3

0.40 pts

Find n .

You can calculate coefficients $\varkappa_{||,0}$ and n using least squares for a modeling function of the form:

$$\varkappa_{||} = \frac{\varkappa_{||,0}}{T_0^n}$$

using a standard calculator.

Answer:

$$n = 1.03$$

B4

1.40 pts

Follow steps **B2** and **B3** for tungsten sulfide (WS_2).

$T_0 = 273$ K. $\bar{d}$, μm 120 140 160 180 200 160 ΔT , K 1.596 1.417 1.284 1.163 1.084 1.309 $\Delta T d^{-1}$, 10^{-3} K $\cdot \mu\text{m}^{-1}$ 13.30 10.12 8.03 6.46 5.42 8.67 $\Delta T d^{-2}$, 10^{-6} K $\cdot \mu\text{m}^{-2}$ 110.83 72.30 50.16 35.90 27.10 59.26
 d^{-2} , 10^{-6} μm^{-2} 69.44 51.02 39.06 30.86 25.00 43.08 d^{-3} , 10^{-9} μm^{-3} 578.70 364.43 244.14 171.47 125.00
296.75 d^{-4} , 10^{-9} μm^{-4} 4.823 2.603 1.526 0.953 0.625 2.106 $T_0 = 323$ K $\bar{d}$, μm 120 140 160 180 200
160 ΔT , K 1.678 1.482 1.370 1.255 1.162 1.389 $\Delta T d^{-1}$, 10^{-3} K $\cdot \mu\text{m}^{-1}$ 13.98 10.59 8.56 6.97 5.81 9.18
 $\Delta T d^{-2}$, 10^{-6} K $\cdot \mu\text{m}^{-2}$ 116.53 75.61 53.52 38.73 29.05 62.69 d^{-2} , 10^{-6} μm^{-2} 69.44 51.02 39.06 30.86
25.00 43.08 d^{-3} , 10^{-9} μm^{-3} 578.70 364.43 244.14 171.47 125.00 296.75 d^{-4} , 10^{-9} μm^{-4} 4.823 2.603 1.526
0.953 0.625 2.106 $T_0 = 373$ K $\bar{d}$, μm 120 140 160 180 200 160 ΔT , K 1.763 1.607 1.509 1.355 1.307 1.508
 $\Delta T d^{-1}$, 10^{-3} K $\cdot \mu\text{m}^{-1}$ 14.69 11.48 9.43 7.53 6.54 9.93 $\Delta T d^{-2}$, 10^{-6} K $\cdot \mu\text{m}^{-2}$ 122.43 81.99 58.95 41.82
32.68 67.57 d^{-2} , 10^{-6} μm^{-2} 69.44 51.02 39.06 30.86 25.00 43.08 d^{-3} , 10^{-9} μm^{-3} 578.70 364.43 244.14
171.47 125.00 296.75 d^{-4} , 10^{-9} μm^{-4} 4.823 2.603 1.526 0.953 0.625 2.106 You can calculate coefficients
 $\varkappa_{||,0}$ and n using least squares for a modeling function of the form:

$$\varkappa_{||} = \frac{\varkappa_{||,0}}{T_0^n}$$

using a standard calculator.

You can calculate coefficients $\varkappa_{||,0}$ and n using least squares for a modeling function of the form:

$$\varkappa_{||} = \frac{\varkappa_{||,0}}{T_0^n}$$

using a standard calculator.

$$\rho = 1.000 \cdot 10^3 n = 1.02$$

Answer: T_0 , K 273 323 373 A, $\mu\text{m} \cdot \text{K}$ 250.0 276.7 325.7 $\times$, $\frac{\text{W}}{\text{m} \cdot \text{K}}$ 37.7 34.1 29.0

$$\rho = 1.000 \cdot 10^3 n = 1.02$$

B5

0.60 pts

Using the least squares method, obtain a formula for finding the coefficient $\Delta T_{\max}$ for each temperature value T_0 and for each material. Express your answer in terms of sample average values of products of the form $\Delta T^{k_1} d^{k_2}$, where $k_{1,2}$ are some numbers.

According to the method described in the condition, we compose a system of equations:

$$\begin{cases} \frac{\partial S}{\partial (\Delta T_{\max})} = 0 \\ \frac{\partial S}{\partial C} = 0 \end{cases}, \quad \text{where} \quad S = \sum_i (\Delta T_{i \text{ exp}} - \Delta T_{i \text{ th}}(d_i, A, B))^2 = \sum_i (\Delta T_i - \Delta T_{\max} + C d_i)^2$$

Thus, the system will take the form:

$$\begin{cases} \frac{\partial}{\partial (\Delta T_{\max})} \sum_i (\Delta T_i - \Delta T_{\max} + C d_i)^2 = 0 \\ \frac{\partial}{\partial C} \sum_i (\Delta T_i - \Delta T_{\max} + C d_i)^2 = 0 \end{cases} \Rightarrow \begin{cases} 0 = \sum_i (\Delta T_i - \Delta T_{\max} + C d_i) \\ 0 = \sum_i (\Delta T_i d_i - \Delta T_{\max} d_i + C d_i^2) \end{cases}$$

This is a system of linear equations for the coefficients A and B . The coefficients in it are necessary products of the form $\Delta T^{k_1} d^{k_2}$. Using the average values of these products over the experimental sample, this system is written somewhat more elegantly in the form:

$$\begin{cases} 0 = \overline{\Delta T} - \Delta T_{\max} + C \bar{d} \\ 0 = \overline{\Delta T d} - \Delta T_{\max} \bar{d} + C \bar{d}^2 \end{cases}$$

Solving this system for $\Delta T_{\max}$, we finally obtain:

$$\Delta T_{\max} = \frac{\overline{\Delta T} \cdot \bar{d}^2 - \bar{d} \cdot \overline{\Delta T d}}{\bar{d}^2 - \bar{d}^2}.$$

Answer:

$$\Delta T_{\max} = \frac{\overline{\Delta T} \cdot \bar{d}^2 - \bar{d} \cdot \overline{\Delta T d}}{\bar{d}^2 - \bar{d}^2}$$

B6

1.20 pts

For each of the materials, find $\Delta T_{\max}$ for each of the temperatures T_0 .

You can, of course, use the formula from the previous paragraph, but since the modeling function is simply a linear relationship, the value $\Delta T_{\max}$ can be calculated using least squares and using a standard calculator.

Answer: T_0 , K 273 323 373 $\Delta T_{\max}^{\text{MoS}_2}$, K 2.841 2.451 2.159 $\Delta T_{\max}^{\text{WS}_2}$, K 3.155 2.679 2.293

B7

0.80 pts

For each of the materials, find m . What makes the main contribution to heat transfer between the sample and the environment – thermal conductivity or radiation? Underline the correct option on your answer sheet.

You can calculate coefficients $\Delta T_{\max,0}$ and m using least squares for a modeling function of the form:

$$\Delta T_{\max} = \frac{\Delta T_{\max,0}}{T_0^m}$$

using a standard calculator. As a result, we obtain: for MoS_2 , $m = 0.88$ – the main contribution is made by *thermalconductivity* / radiation; for WS_2 , $m = 0.83$ – the main contribution is made by *thermalconductivity* / radiation.

You can calculate coefficients $\Delta T_{\max,0}$ and m using least squares for a modeling function of the form:

$$\Delta T_{\max} = \frac{\Delta T_{\max,0}}{T_0^m}$$

using a standard calculator. As a result we get:

The main contribution to heat transfer between the sample and the environment is thermal conductivity.

Answer: Material MoS_2 WS_2 m 0.88 0.83 Thermal conductivity makes the main contribution to the heat transfer between the sample and the environment.